



Numerical Analysis of the Aerodynamic and Vibration Performance of High-Speed Elevators Using Air-Structure Interaction

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Received: 01-05-2026

Revised: 02-25-2026

Accepted: 03-19-2026

Citation: Z. M. Agool, D. Alkhafaji, and M. J. Aubad, "Numerical analysis of the aerodynamic and vibration performance of high-speed elevators using air-structure interaction," *Int. J. Comput. Methods Exp. Meas.*, vol. 14, no. 1, pp. 125–140, 2026. <https://doi.org/10.56578/ijcmem140108>.



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Abstract: This research investigates the aerodynamic performance and dynamic response of high-speed elevators. The study was conducted using a numerical model based on a two-way air-structure coupling. This is achieved by integrating computational fluid dynamics (CFD) and finite element analysis (FEA) techniques. Three different elevator cabin designs (flat, elliptical, and dome) were analyzed at different operating speeds (6, 8, 10, and 12 m/s) to evaluate the effect of geometry on flow and vibration characteristics. The results showed that the dome cabin shape achieved the best overall performance, contributing to reductions of approximately 41% in acceleration, 35% in deformation, 28% in stress, and the vibration frequency by approximately 50–60% compared to the flat shape. It also exhibited a significant reduction in vibration amplitude. Furthermore, a critical dynamic amplification region was identified at approximately 10 m/s, where the response reaches its peak. This region should be considered when designing damping systems. This improvement is attributed to the streamlined properties of the cabin's dome shape, which reduce flow decoupling and pressure fluctuations. The results show that improving the streamlined shape may reduce air resistance, thereby positively impacting the required operating power.

Keywords: High-speed elevator; Aerodynamic characteristics; Vibration; Gas-solid coupling system; Fluid-structure interaction

1 Introduction

Elevators are a means of transportation that have evolved alongside the construction of high-rise buildings, and their status is comparable to that of vertically moving cars. In high-rise buildings, there is an increasing demand for high-speed elevators to reduce congestion, save passengers' commuting time, and improve elevator efficiency. Therefore, as the elevator's operating speed increases, its vibrations become more pronounced. This increased vibration is due to the interaction of the airflow field with the cabin structure. Consequently, as the elevator's operating speed increases, the coupling effect of the airflow in the shaft with the cabin structure becomes more significant. A strong vibration will not only damage some key parts of the elevator system but also reduce its service life, and seriously affect the stability, safety, and comfort of the elevator [1]. Despite the growing interest in high-speed elevators, studies that comprehensively address aerodynamic performance and vibrational response remain limited. Most recent research has focused on improving aerodynamic properties to reduce air resistance or noise, enhance passenger comfort, and improve operating efficiency, without directly addressing the system's dynamic and structural behavior.

In the field of geometry and its optimization, studies by Zhang et al. [2, 3] have shown that using streamlined shapes, such as toe guards or curved covers, significantly reduce drag, minimize back vortices, and improve pressure distribution. These studies also showed that improving the radius of curvature enhances lateral stability and reduces high-pressure zones, reflecting the influence of geometry on flow behavior. However, these analyses remained confined to the aerodynamic framework, without assessing their impact on vibrational response or dynamic loads. Similarly, numerical studies such as those by Su et al. [4] have employed multi-objective techniques to improve aerodynamic performance, resulting in enhanced cabin stability, reduced lateral torque, lower drag coefficient, and reduced energy consumption. Studies addressing unsteady flow and noise, such as those by He et al. [5] and Zhang et al. [6], have confirmed that vortex formation and flow detachment are the primary sources of noise and aerodynamic

turbulence, and that geometry control can significantly reduce these effects. However, these phenomena have not been directly linked to vibration behavior or dynamic amplification.

Regarding the interaction between the counterweight and the cabin, Zhang et al. [7] demonstrated that this interaction significantly and directly affects the lifting and compressive forces and operational stability, particularly at intersection moments. However, this effect on forces and its impact on vibrational response have not been adequately analyzed. Some studies indicate that the aerodynamic behavior inside an elevator shaft is directly affected by multiple engineering and operational factors. Cui et al. [8] demonstrated that increasing the blockage ratio within the shaft increases resistance to motion and flow turbulence, especially in the rear region of the cabin, thereby enhancing flow instability. Similarly, Wang et al. [9] showed that reducing the blockage ratio reduces drag and pressure loads, thereby improving aerodynamic performance and mitigating adverse effects on elevator stability. Qiao et al. [10] also presented a theoretical analysis of the piston effect, explaining that cabin motion within a narrow shaft generates unstable flow that intensifies with velocity, contributing to increased noise and aerodynamic loads on the cabin. On the other hand, Yang et al. [11] focused their study on the effect of the distance between the cabin and the well wall, where the results showed that reducing this distance leads to an increase in the intensity of the flow and an increase in aerodynamic forces, which negatively affect the vibrations and noise.

Recent studies have also shown that energy consumption in high-speed elevators is directly related to the increased air resistance caused by cabin movement within the shaft, with consumption increasing non-linearly with increasing operating speed. Chen et al. [12] demonstrated that adding deflectors reduces air resistance, leading to a significant decrease in the energy required to operate the elevator, especially at high speeds, with multi-surface aerodynamic shapes achieving optimal performance. Similarly, several studies have developed advanced strategies to improve the energy efficiency of elevator systems. Hu et al. [13] developed a multi-objective genetic algorithm (MOGA) for elevator scheduling that balances energy consumption and passenger waiting times, yielding energy savings of up to 23.6% compared to conventional methods. Zhang and Zong [14, 15] proposed an advanced model for optimizing energy distribution during peak periods, based on scheduling that accounts for traffic flow uncertainties, thereby improving operational efficiency and reducing consumption. Furthermore, Desdowitz et al. [16] developed a dual-control system for energy storage management, based on an advanced and a low-level controller, demonstrating the potential for energy savings of up to 35%. Zhang et al. [17] also presented a DC microgrid-based approach that achieved energy efficiencies ranging from 15.87% to 23.1% in experimental tests and 54.5% in field applications, reflecting the significant potential of these technologies to improve elevator system efficiency.

Despite these efforts, most studies have focused on improving energy consumption from the perspective of control systems or energy management, without directly addressing aerodynamic effects and the dynamic interaction between airflow and the elevator structure.

On the other hand, dynamical studies such as those by Feng et al. [18] and Zhang et al. [19, 20] have focused on simulating vibrations induced by excitation of the guide rails. Models with different degrees of freedom have been studied to analyze lateral vibration and the influence of rail characteristics. Also, Qiu et al. [21], the vibration behavior in high-speed elevators is primarily related to system characteristics such as stiffness, damping, and excitation frequencies. Their proximity to the natural frequency amplifies vibrations, leading to resonance. It has also been shown that certain design parameters, such as the characteristics of the steering system, directly influence the level of vibrational acceleration. However, these models do not rely on a representation of aerodynamic forces and do not reflect the true interaction between airflow and the elevator structure. Whilst Liu et al. [22] and Yang et al. [23] have also incorporated aerodynamic force effects into vibration models. However, these models have relied on a simplistic representation of aerodynamic forces, treating them as static or unidirectional, which fails to reflect the true interaction between the flow and the structure. The concept of two-way coupling has been widely applied in other fields, such as vibrations caused by flow [24], wind turbine blades [25], and in studies of cylindrical structures [26], but its application in high-speed elevators remains limited.

Despite advances in aerodynamics, energy consumption, and dynamic modeling of high-speed elevators, most research has addressed these aspects in isolation. This has highlighted a lack of models capable of accurately representing airflow interactions with the elevator structure, especially at high speeds.

Accordingly, this research aims to develop an integrated numerical model of fluid-structure interaction (FSI) to link aerodynamic behavior to dynamic response. It analyzes the effect of geometry on pressure, forces, acceleration, deformation, stress, and frequency response. It provides a quantitative evaluation to identify the most efficient geometry for reducing loads and vibrations and improving energy efficiency.

2 Methodology and Numerical Method

This research is based on an integrated numerical analysis combining aerodynamic modeling and vibration analysis within the framework of FSI simulation. This approach aims to understand the interplay between airflow in an elevator shaft and the cabin's dynamic response during high-speed operation. A quantitative research methodology based on computational modeling was adopted, using advanced computational fluid dynamics (CFD) and finite

element analysis (FEA) simulation tools to analyze the unsteady behavior of air and its interaction with the elevator structure.

2.1 Geometric Model

The considerable complexity of elevator system components in actual operation makes building a simulation model more difficult and may limit the generalizability of the analysis results. Since the study of air-structure interaction focuses on the cabin wall boundaries, a systematic simplification of the physical model was adopted to align with real-world operating conditions, as follows:

1. The high-speed elevator cabin system was considered a homogeneous mass, which could affect the distribution of local stresses without altering the overall direction for comparison between different shapes. The study used three different cabin shapes, as shown in Figure 1, to illustrate the effect of cabin shape, neglecting the complex internal and external connection structure of the cabin system.

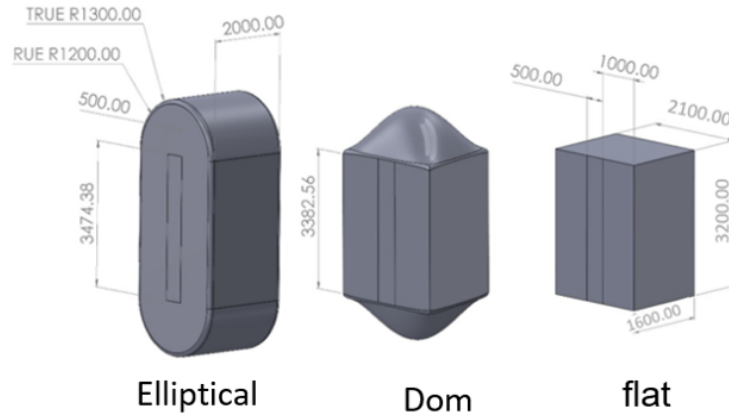


Figure 1. Cabin shape models

2. The high-speed elevator shaft was sufficiently long, and the airflow at the far end of the shaft had a minimal effect on the cabin body. Therefore, the effect of the openings at the ends of the shaft on the airflow in the central section, which operates at a constant speed, can be disregarded. While they may cause limited air turbulence, especially at the ends, their effect in the study area (the middle of the shaft) remains minimal.

3. The airflow within the well caused by temperature differences is limited compared to the operating speeds of high-speed elevators. Therefore, the chimney effect can be justifiably neglected, given its negligible impact on the airflow generated by the rapid movement. Thus, the results of this study can reflect the assumed operating conditions.

To minimize the influence of other factors on the results of this study's analysis, several factors were kept constant, including the elevator cabin, elevator shaft, blocking ratio, and the distance between the cabin and the shaft wall. Figure 2 shows a simplified three-dimensional geometric model of the shaft and cabin flow field. At the same time, Table 1 summarizes all the geometric parameters and constant dimensions of the full-size reference cabin.

Table 1. Standard cabin specifications used in numerical analysis

Parameter	Symbol	Value	Notes
Shift length (travel axis)	L_h	2.91 m	Clear internal shift length ≈ 2.9 m
Shift width (lateral)	W_h	2.51 m	Clear internal shift width ≈ 2.5 m
Shift height	H_h	1.48 m	Clear internal shift height ≈ 148 m
Cabin length (travel axis)	L	2.10 m	Clear internal car length ≈ 2.0 m + allowances
Cabin width (lateral)	W	1.70 m	Clear internal car width ≈ 1.6 m + allowances
Cabin height (clear)	H	3.2 m	Overall height to roof skin ≈ 3.50 m
Wall thickness	t_w	3.0 mm	Steel sheet + stiffeners
Floor plate thickness	t_f	5.0 mm	Laminated plate over beams
Roof plate thickness	t_r	3.0 mm	With ventilation cutouts
Door opening	-	1.00×2.10 m	Centered on Y-Z plane
Guide-rail spacing	s_r	1.90 m	Rail centers in X direction
Rail offset from corners	O_r	100 mm	From cabin outer shell
Corner fillet radius	R_c	300 mm	Plan-view fillet radius
Dome curvature radius	R_d	1.20 m	Hemisphere cap blended at roof

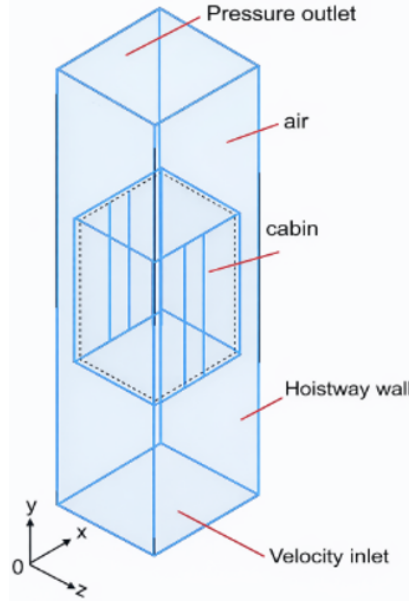


Figure 2. The 3D geometry model of the flow field and the cabin body

2.2 Governing Equations

The mathematical model of the system is based on a set of fundamental laws, including Newton's second law of motion and the theory of linear elasticity. These frameworks describe the aerodynamic behavior of the air inside and around the elevator shaft as the cabin moves at high speeds. The flow is assumed to be turbulent and incompressible [27]. The continuity equation:

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u_i)}{\partial x_i} = 0 \quad (1)$$

where, ρ represents the air density (kg/m^3), u_i represents the components of the velocity vector (m/s), and x_i represents the spatial coordinates.

Three-dimensional Navier–Stokes equation:

$$\frac{\partial (\rho u_i)}{\partial t} + \frac{\partial (u_i u_j)}{\partial x_j} = \rho f_i + \frac{\partial \sigma_{ij}}{\partial x_j} \quad (2)$$

where, f_i represents the forces exerted on the body (N/kg), σ_{ij} represents the fluid stress components (Pa), and u_j represents the velocity component in the direction j .

Due to the unstable nature of the airflow in the elevator shaft, the turbulence model chosen was the RNG $k - \epsilon$ model, a model commonly used in engineering, which incorporates equations for the transfer of turbulent kinetic energy and its dissipation rate, taking into account the vibration acceleration for the six degrees of freedom of the cabin wall boundary in the flow field.

The equation for disturbance energy is:

$$\frac{\partial K}{\partial t} + \bar{u}_i \frac{\partial K}{\partial x_i} = \frac{\partial}{\partial x_i} \left[\left(C_K \frac{K^2}{\epsilon} + \nu \right) \frac{\partial K}{\partial x_i} \right] + P - \epsilon \quad (3)$$

The dissipation rate equation is:

$$\frac{\partial \epsilon}{\partial t} + \bar{u}_i \frac{\partial \epsilon}{\partial x_i} = \frac{\partial}{\partial x_i} \left[\left(C_\epsilon \frac{K^2}{\epsilon} + \nu \right) \frac{\partial \epsilon}{\partial x_i} \right] + C_{\epsilon 1} \frac{\epsilon}{K} P - C_{\epsilon 2} \frac{\epsilon^2}{K} \quad (4)$$

where, k is the kinetic energy of the disturbance (m^2/s^2), ϵ is its dissipation rate (m^2/s^3), ν is the kinetic viscosity, and P is the disturbance production rate. $P = \nu_t \left(\frac{\partial \bar{u}_i}{\partial x_k} + \frac{\partial \bar{u}_k}{\partial x_i} \right) \frac{\partial \bar{u}_i}{\partial x_k}$. $C_K, C_\epsilon, C_{\epsilon 1}, C_{\epsilon 2}$ are empirical coefficients (adjusted through experimental data and simulation results in the literature [28]).

Dynamic equation of motion:

$$M\ddot{u}(t) + C\dot{u}(t) + Ku(t) = F_{\text{aero}}(t) + F_{\text{inertial}}(t) \quad (5)$$

where, M is the mass matrix, C is the damping matrix, K is the stiffness matrix, $u(t)$ is the displacement, F_{aero} is the aerodynamic forces, and F_{inertial} is the Inertial forces. Eq. (5) describes the relationship between mass M , damping C , and stiffness K with displacement $u(t)$ under the influence of aerodynamic and dynamic forces [29].

The relationship between stress and strain is represented by Hooke's law for homogeneous linear materials:

$$\sigma = D\varepsilon \quad (6)$$

where, σ is the stress vector, ε is the strain vector, and D is the elastic matrix, given for isotropic materials as:

$$D = \frac{E}{(1+v)(1-2v)} \begin{bmatrix} 1-v & v & v & 0 & 0 & 0 \\ v & 1-v & v & 0 & 0 & 0 \\ v & v & 1-v & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1-2v}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1-2v}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1-2v}{2} \end{bmatrix} \quad (7)$$

where, E is Young's modulus, and v is Poisson's ratio.

2.3 Meshing and Boundary Conditions

The mesh generation process is a crucial step in numerical setup, as it directly impacts the accuracy of the results and the stability of the solution. In this study, an irregular mesh of tetrahedral elements was used to represent the air space surrounding the compartment. Near-surface amplification layers were added to enhance the representation of the flow behavior. Furthermore, mesh independence testing was performed to ensure the results' reliability. Three progressively improved grids were developed for each elevator geometry: a coarse mesh (approximately 45,000 elements), a medium mesh (approximately 120,000 elements), and a fine mesh (approximately 220,000 elements). Key performance indicators, such as maximum strain, peak deformation, and dominant frequency response amplitude, were collected from the respective grids. Table 2 shows the parameters for each of the three grid combinations. Figure 3 illustrates the mesh configuration near the car wall and the column wall for the mesh densities used in this study. The torsion coefficient, aspect ratio, and Jacobian determinant of elements were examined for all grids. Over 95% of the elements met the proposed criteria (torsion coefficient < 0.85 , aspect ratio < 5 , Jacobian determinant > 0.6), indicating that the segmentation process was free of numerical instability. The results were similar across cases because the segmentation methodology was uniform across all three elevator shapes (flat, elliptical, and dome).

Table 2. Mesh independence study results

Mesh Type	Elements	Max Stress (Pa)	Max Deformation (m)	Peak Response	Relative Difference
Coarse	~45,000	1.23×10^7	6.1×10^{-4}	1.12×10^5	6–9%
Medium	~120,000	1.18×10^7	5.9×10^{-4}	1.10×10^5	$< 2\%$
Fine	~220,000	1.17×10^7	5.8×10^{-4}	1.09×10^5	$< 1.68\%$

To assess mesh independence, the results of the coarse and medium meshes were compared with those of the fine mesh, which served as the reference case. The assessment was based on calculating the percentage difference by measuring the deviation of each parameter from its value in the fine mesh.

$$\text{Percentage Difference} = \left| \frac{X_{\text{mesh}} - X_{\text{fine}}}{X_{\text{fine}}} \right| \times 100 \quad (8)$$

where, X_{mesh} is the value obtained from the coarse or medium mesh, and X_{fine} is the value obtained from the fine mesh.

In this study, the middle uniform-speed section of the elevator was used as the simulation object. Based on the concept of a wind-tunnel test, the car body was stationary within the shaft, and airflow was directed at the cabin at the actual elevator operating speed. The simulation method only affected the relative motion between the airflow and the shaft wall, as well as the interaction between the airflow in the shaft and the cabin body [30]. In addition, the boundary-condition settings for the numerical simulation domain are shown in Figure 2. The bottom of the flow field was set as the velocity inlet, and the top was set as the pressure outlet. This paper selected four operating speeds: 6 m/s, 8 m/s, 10 m/s, and 12 m/s. A spring-smoothing method and a meshing method were used to reconstruct and update the cabin-body boundary grid at each time step. Additionally, the cabin wall was set as the System Coupling surface for a gas-solid, used to transfer aerodynamic force data from the shaft flow to the cabin body calculation domain. In addition to the fluid boundary conditions, the structural constraints of the cabin were defined to reflect guide-rail interactions, where lateral motion was restricted, longitudinal motion was allowed, and prescribed excitations governed vertical motion.

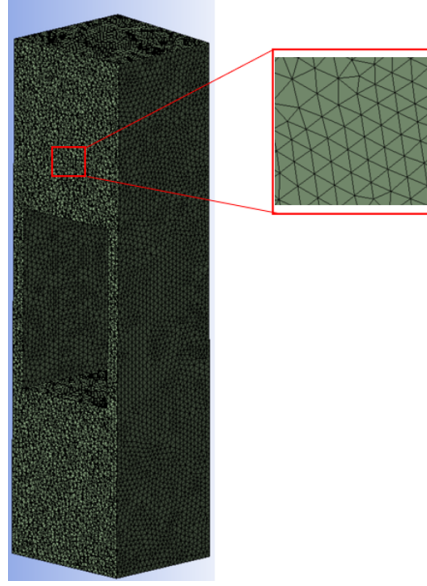


Figure 3. Surface grid of mesh

2.4 Time Step Sensitivity Test

A time-step sensitivity test was conducted to assess the stability and accuracy of the numerical solution in unsteady simulations. For this purpose, three different time step values were adopted: 5×10^{-4} seconds, 2×10^{-4} seconds, and 1×10^{-4} seconds. The smallest time-step value was used as the reference case for comparison. To evaluate the effect of changing the time step, the relative difference between the results obtained from each case and the reference result was calculated using the following relationship:

$$\text{Error (\%)} = \left| \frac{P_{\Delta t} - P_{\text{ref}}}{P_{\text{ref}}} \right| \times 100 \quad (9)$$

where, $P_{\Delta t}$ represents the pressure value obtained at a given time step, and P_{ref} represents the pressure value calculated at the smallest time step. The results showed that the differences between the studied cases remained within narrow limits and did not exceed (2%), which indicates that the time step that was adopted provides an appropriate level of accuracy while maintaining the efficiency of the calculations.

This methodology for testing time independence is consistent with similar studies in unsteady flow and air-structure interaction analysis, such as Han et al. [31], which studied the aeroelastic response of a high-speed elevator cabin.

2.5 Air-Structure Coupling

A two-way interactive coupling between the aerodynamic and structural domains was employed to represent the interaction between the flow within the elevator shaft and the cabin response under aerodynamic loads. The coupling was performed using numerical analysis in ANSYS Workbench, where CFD was combined with FEA under unsteady-state conditions. Figure 4 illustrates the coupling process flow, showing the data exchange sequence between the two domains and the iteration pattern within each time step. As the diagram shows, the calculations were performed in a segmented and simultaneous manner to ensure the stability of the numerical solution.

The coupling process at each time step begins by calculating the pressure distribution on the cabin surfaces using a CFD model. These values are then transferred to the FEA model and applied as surface loads at the interaction interface. The FEA model calculates the resulting displacements and deformations. The calculated displacements are then fed back into the airspace, where they are used to update the computational mesh via dynamic meshing. This step represents the essential link between the two domains, as structural deformations directly affect the flow characteristics, as illustrated in Figure 4. The data exchange between the flow analyzer and the structural analyzer is repeated within the same time step until the convergence condition is met. An upper limit is set on the number of coupling iterations per time step to ensure solution stability, with a convergence criterion that minimizes the change in values transmitted between the two domains. Furthermore, the cabin motion is represented by introducing a base motion into the structural model, while the inlet flow velocity is defined as equal to the elevator's operating speed. This representation ensures that the relative velocity between the air and the cabin is accurately represented without requiring a complete computation of the computational domain.

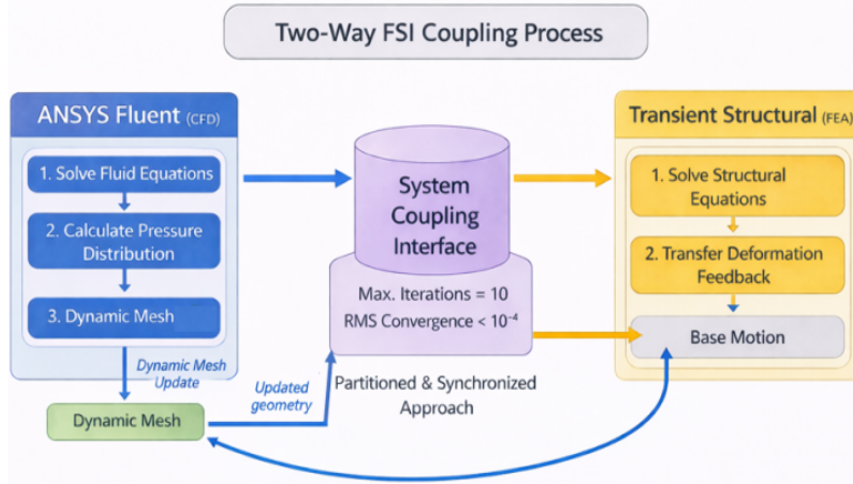


Figure 4. A diagram illustrating the bidirectional coupling mechanism between the aerodynamic and structural domains

Consequently, this approach provides a comprehensive representation of the interaction between air loads and structural response, enabling precise analysis of vibration, stress, and deformation behavior under unstable operating conditions.

2.6 Numerical Simulations

ANSYS Workbench 2022R2 was used to model the transient structural and dynamic response of an elevator cabin at various shapes and speeds. Structural analysis was also combined with fluid dynamics to simulate the air pressure distribution over a transient timeframe encompassing the acceleration, stabilization, and deceleration phases.

Models were imported in Parasolid (x_t) format and geometrically processed. A two-way FSI was used, in which flow stresses computed by CFD were transferred to the structural model to calculate aerodynamic forces and the resulting deformations. Loading conditions included aerodynamic forces, base motion along a trapezoidal velocity curve, and gravity, with concentrated masses added to represent realistic inertia. Main outputs included acceleration, deformation, von Mises stresses, and frequency response, with numerical solution convergence ensured within limits ($<10^{-5}$ for fluids and $<10^{-6}$ for structures).

2.7 Modeling Properties of Simulations

The numerical model employs two-way air-structure coupling, using CFD and FEA to represent the interaction between airflow and cabin response accurately. Linear elastic steel with Rayleigh damping is employed. The boundary conditions were designed to represent realistic operating constraints and air movement within the elevator shaft. The B.C were determined according to the different degrees of freedom. Lateral movement (X) is fully restricted, while movement in the direction of travel (Y) is free. In the vertical direction (Z), a displacement representing the elevator's movement is imposed. Rotation at the guide rail contact points is also restricted to ensure a realistic representation of the support. Load conditions include aerodynamic forces, gravity, and basal excitation along a trapezoidal velocity curve, with mass representation using a combination of structural and concentrating masses. This formulation achieves an effective balance between physical accuracy and computational efficiency. To ensure full clarity and reproducibility of the structural model, all modeling parameters, including material properties, damping coefficients, mass distribution, and boundary conditions, are summarized in Table 3.

3 Results and Discussion

This section provides a detailed analysis of numerical simulations of various cabin models in a high-speed elevator shaft, evaluating their aerodynamic, dynamic, and vibrational performance under different operating conditions. The results are presented in sections covering key system performance indicators: airflow behavior analysis, pressure distribution, acceleration, deformation, stress distribution, and frequency response in the X- and Z-directions. The results are compared across three geometries (flat, elliptical, and dome), with a particular focus on the effects of variable operating speeds (6, 8, 10, and 12 m/s) on system stability and safety.

3.1 Analysis of Elevator Velocity Profile at Different Operating Speeds

Figure 5 shows the evolution of speed over time at different operating speeds (6, 8, 10, 12 m/s). The behavior appears in three main phases: acceleration, then constant speed, then deceleration. As the maximum speed increases,

the journey time decreases significantly, but this higher acceleration rate increases the impact of dynamic loads on the system. This indicates the need to strike a balance between transport efficiency and operational comfort.

Table 3. Modeling and simulation properties summary

Category	Description
Material properties	Inear elastic, homogeneous, isotropic steel; Young's modulus ($E = 210 \text{ GPa}$), Poisson's ratio ($\nu = 0.3$), density ($\rho = 7850 \text{ kg/m}^3$).
Damping assumptions	Rayleigh damping model ($C = \alpha M + \beta K$), were selected, Mass-proportional coefficient (α) = 0.021 1/s and Stiffness-proportional coefficient (β) = $1 \times 10^{-5} \text{ s}$.
Boundary conditions	Rail supports with constrained lateral motion (X), free motion in travel direction (Y), and controlled vertical response (Z); aerodynamic boundary conditions include velocity inlet (equal to elevator speed), pressure outlet.
Mass distribution	Structural mass of steel cabin (950kg) + lumped masses representing doors (120kg), a roof and attachments (80kg), internal equipment (100kg), total modeled mass (1250 kg).
Loading conditions	Combined aerodynamic forces from computational fluid dynamics (CFD) mapping, gravity load (-Z direction) $g = -9.81 \text{ m/s}^2$, dynamic base excitation using trapezoidal velocity profile and operating velocity (6, 8, 10, 12 m/s).
Modeling assumptions	Small deformation theory, perfect connections (no slip), no thermal effects, no passengers (loads simplified as lumped masses), airflow modeled using steady inlet velocity.

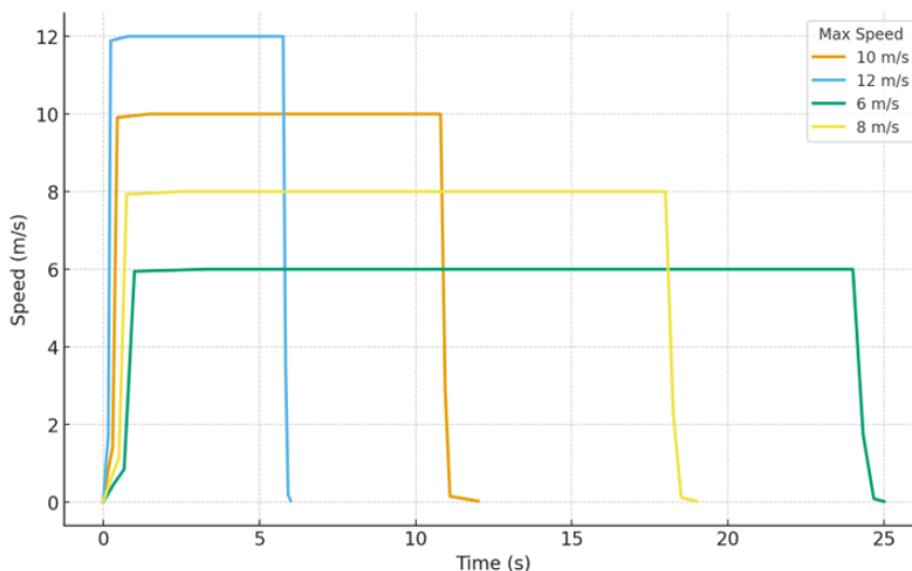


Figure 5. Elevator velocity profiles at different maximum speeds

3.2 Analysis of Pressure Distribution Within the Elevator Shaft

Pressure distribution within an elevator shaft is a key indicator of cabin aerodynamic performance in high-speed systems. CFD analysis was used to study the effect of travel speed (6, 8, 10, 12 m/s) and cabin shape (flat, elliptical, dome) on the flow pattern within the shaft.

Figure 6 shows that the pressure distribution is affected by increasing speed, with low-pressure areas behind the compartment becoming more pronounced, especially at high speeds. The flat shape exhibits the largest pressure difference between its front and rear, indicating increased flow separation. In contrast, the domed shape features a more balanced pressure distribution with a marked decrease in low-pressure areas, indicating smoother flow and less turbulence. The elliptical shape exhibits intermediate behavior between the two.

3.3 The Effect of Elevator Shape

Regarding the dynamic response, Figure 7 shows that the flat cabin registers the highest acceleration at approximately 0.00022 m/s^2 , followed by the elliptical cabin. In contrast, the dome cabin exhibits the lowest acceleration value at around 0.00013 m/s^2 . This indicates the superiority of the dome shape in reducing the dynamic response.

This behavior is attributed to the reduced oscillating aerodynamic forces resulting from improved flow smoothness around the compartment.

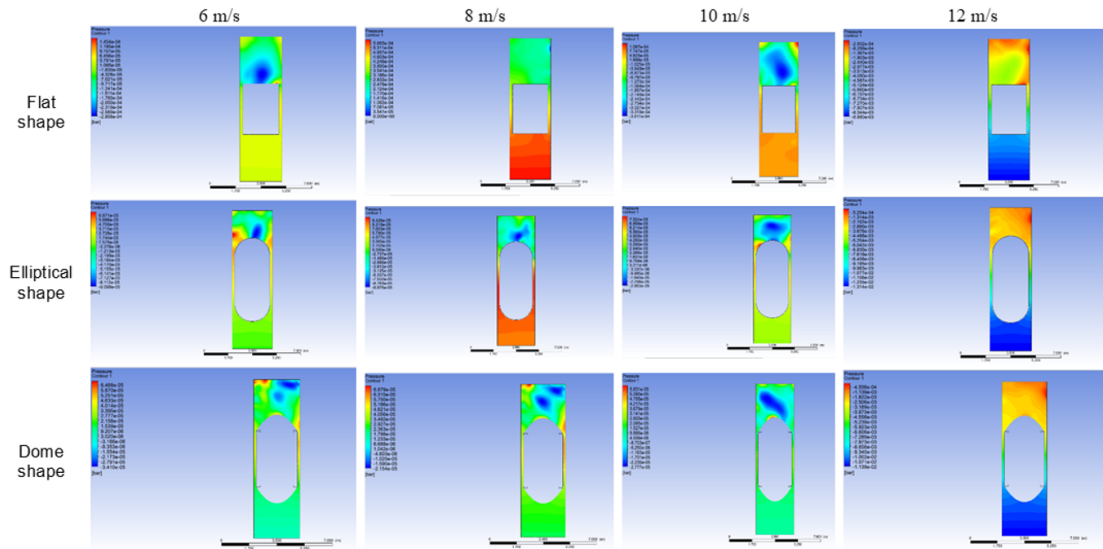


Figure 6. Pressure distribution inside the elevator shaft around the cabin of different shapes at speeds (6, 8, 10, 12 m/s)

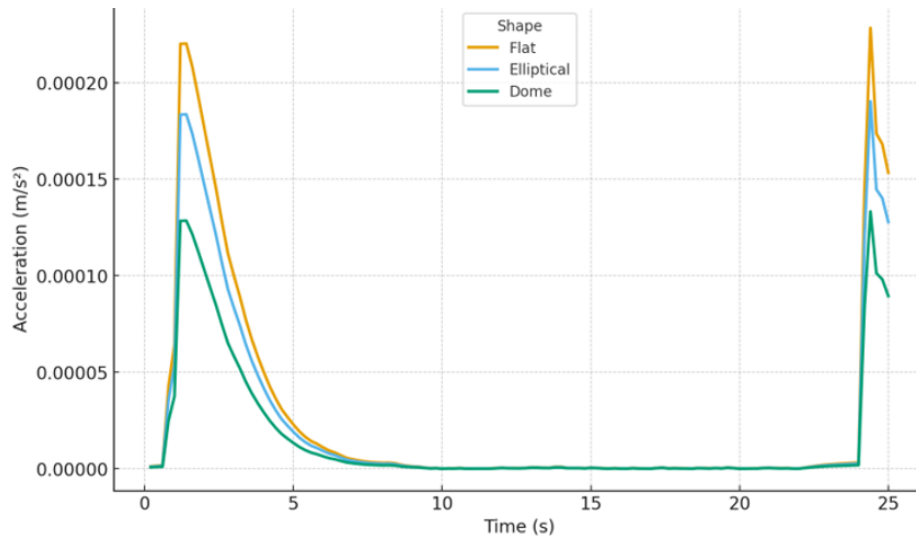


Figure 7. Acceleration vs. time of the different shapes of elevator cabins

Regarding the deformation results in Figure 8, the dome cabin shows the lowest deformation values, which are less than (2.5×10^{-6}) m. The flat cabin registers the highest deformation, reaching (3.8×10^{-6}) m, while the elliptical shape falls in the middle range between the two cases. This indicates a higher structural efficiency for the dome shape. This performance is explained by a more uniform load distribution, as opposed to the heterogeneous pressure distribution in the flat shape.

Similarly, the deformation results in Figure 8 show that the dome shape exhibits the highest structural efficiency, recording the lowest deformation values compared to the other cases, at less than 2.5×10^{-6} m. This is attributed to a more organized load distribution and its impact on stresses. In contrast, the flat shape experiences greater deformation, reaching 3.8×10^{-6} m, due to the uneven pressure distribution. Consequently, the elliptical shape falls in an intermediate position.

Figure 9 illustrates that the maximum stresses are highest in the flat shape, registering approximately 3.05×10^5 Pa, while the dome shape reached a minimum stress of 2.2×10^5 Pa. This demonstrates the influence of the geometric shape on the distribution of internal forces. The dome shape helps reduce stress concentrations by minimizing pressure differentials and improving flow efficiency, thereby enhancing structural integrity and reducing

the likelihood of structural failure.

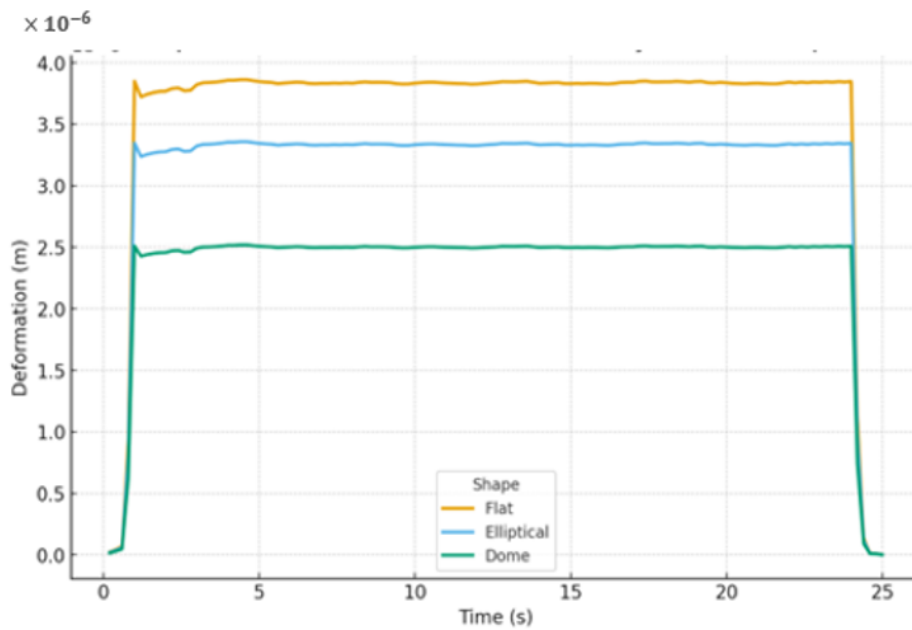


Figure 8. Deformation–time response by elevator shape

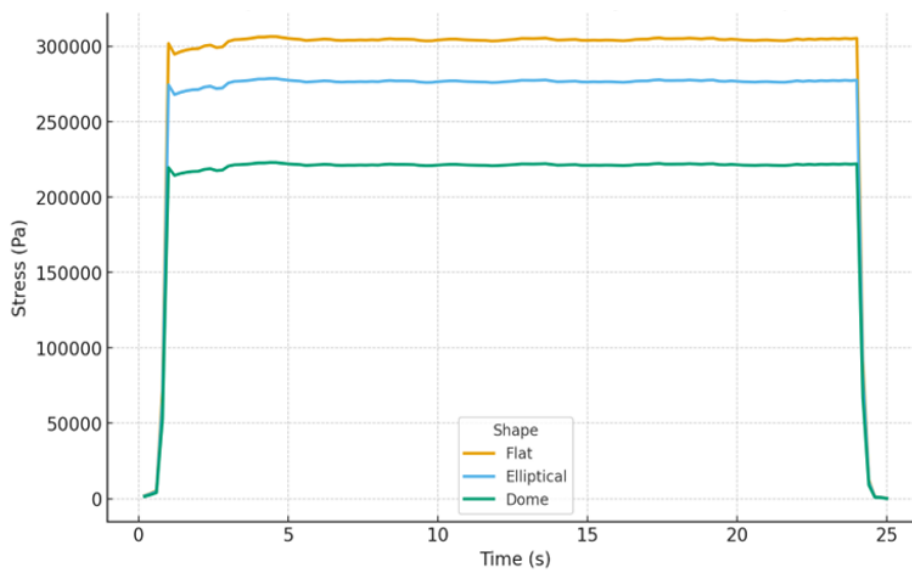


Figure 9. Stress–time response by elevator shape

The frequency response in this study represents the standard structural response amplitude derived from harmonic analysis. It is a relative indicator of vibration intensity and does not express a direct displacement. It is presented on a logarithmic scale to highlight differences across a wide range of frequencies. Figure 10 and Figure 11 show that the horizontal axis represents the frequency (Hz), while the vertical axis represents the response amplitude. The results show that the dome shape exhibits the lowest response levels in the X and Z directions, whereas flat and elliptical shapes achieve higher response levels. It is noteworthy that the horizontal direction is more sensitive to changes than the vertical direction. This performance is attributed to reduced flow detachment and weak vortex formation around the dome shape, which leads to lower aerodynamic excitation. In contrast, the vertical direction exhibits amplification at specific frequencies due to the interaction between flow characteristics and structural response.

The preceding discussion of Figure 7, Figure 8, Figure 9, Figure 10, Figure 11 demonstrates that the cabin geometry directly affects the system's aerodynamic and dynamic responses. To provide a comprehensive evaluation, peak values were extracted for key performance indicators, including acceleration, deformation, stress, and bidirectional frequency response, as shown in Table 4. This allows for a direct comparison between the studied geometries and

the identification of the most efficient one.

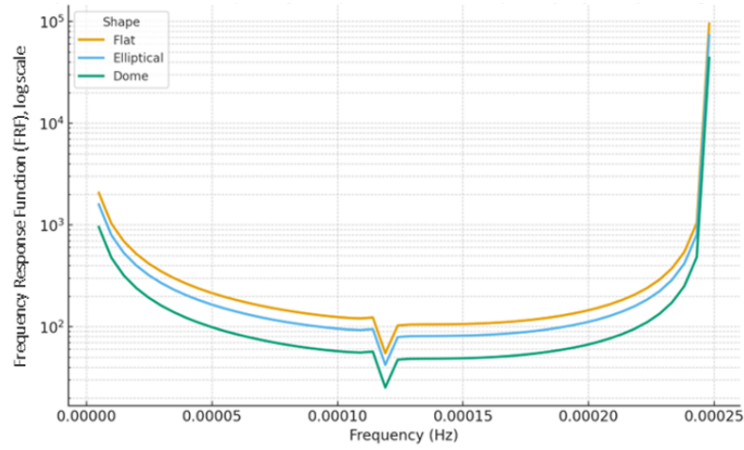


Figure 10. Normalized frequency response amplitude of the elevator cabin in the X-direction for different geometrical shapes vs. frequency, presented on a logarithmic scale

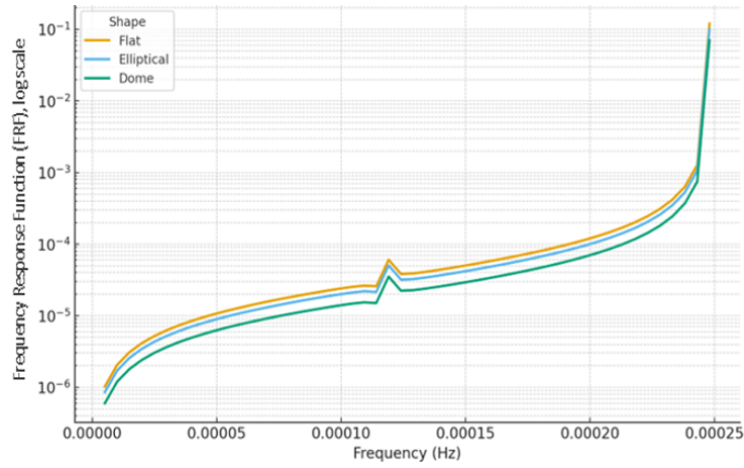


Figure 11. Normalized frequency response amplitude of the elevator cabin in Z-direction for different geometrical shapes vs. frequency, presented on a logarithmic scale

Table 4. Comparative performance of elevator cabin shapes

Parameter	Flat	Elliptical	Dome	Improvement (Dome vs. Flat)
Acceleration (m/s^2)	2.2×10^{-4}	1.8×10^{-4}	1.3×10^{-4}	40.9%
Deformation (m)	3.85×10^{-6}	3.35×10^{-6}	2.50×10^{-6}	35.1%
Stress (Pa)	3.05×10^5	2.75×10^5	2.20×10^5	27.9%
FRF amplitude (X-direction)	$\sim 1 \times 10^5$	$\sim 6 \times 10^4$	$\sim 3 \times 10^4$	$\approx 70\%$
FRF amplitude (Z-direction)	$\sim 1 \times 10^{-1}$	$\sim 6 \times 10^{-2}$	$\sim 4 \times 10^{-2}$	$\approx 60\%$

Note: FRF = Frequency Response Function.

To provide a clear quantitative comparison of the investigated cabin geometries, a relative performance assessment was carried out using the flat configuration as the reference condition. The percentage improvement was calculated as the reduction in each response parameter relative to the baseline. The relationship used for this evaluation is expressed as:

$$\text{Improvement}(\%) = \frac{x_{\text{flat}} - X_{\text{dome}}}{X_{\text{flat}}} \times 100 \quad (10)$$

where, X_{flat} denotes the value obtained for the flat cabin, which is considered as the reference case, while X_{dome} represents the corresponding value for the dome-shaped configuration. All comparisons were based on the peak response values to ensure consistency in the evaluation.

The results in Table 4 show the superiority of the dome shape across all studied performance indicators, as it achieves the lowest values for acceleration, deformation, and stress, and exhibits a significant decrease in frequency response, confirming its aerodynamic and dynamic efficiency compared to the other shapes.

3.4 The Effect of Elevator Speed

Regarding the effect of speed, Figure 12, Figure 13, Figure 14, Figure 15, Figure 16 show that increasing operating speed significantly increases dynamic values. This is attributed to the increased aerodynamic forces acting on the cabin. Figure 12 shows that acceleration increases with speed. Figure 13 shows a gradual increase in deformation with increasing aerodynamic load. Figure 14 shows that stresses increase with speed, reflecting the influence of aerodynamic forces on the structural response.

Regarding the frequency response function (FRF), Figure 15 (horizontal, X-axis) and Figure 16 (vertical, Z-axis) show a clear dynamic amplification at a speed of approximately 10 m/s, where the response attains its highest value compared to other speeds. This increased response indicates a dynamic interaction between the airflow excitation and the system’s structural characteristics at approximately 10 m/s. This behavior can be interpreted as a convergence between the excitation frequency arising from flow instability, such as eddy-current formation, and one of the compartment’s natural frequencies. This convergence leads to an amplified dynamic response. However, the occurrence of resonance cannot be fully confirmed due to the lack of a direct analysis of the natural frequencies. Therefore, this situation can be considered indicative of potential dynamic amplification resulting from FSI. These results highlight the importance of speed, especially at critical intermediate values, and should be taken into account when designing damping systems and improving dynamic stability in high-speed elevators.

In general, the results indicate that optimizing the cabin geometry, particularly adopting a dome shape, reduces air resistance and improves pressure distribution, thereby reducing dynamic loads and vibrations. Therefore, the superiority of the dome shape depends primarily on its overall geometric properties rather than on the specific details of the model, thereby supporting the reliability of the results despite the simplifications employed in this study. This is expected to impact operating efficiency positively; however, assessing energy consumption and operating costs requires a separate, dedicated model.

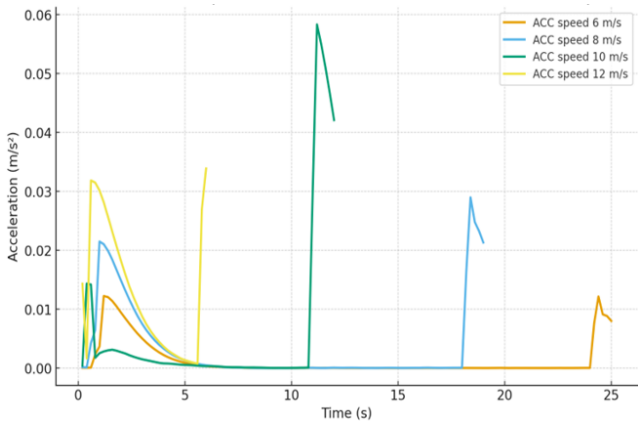


Figure 12. Dome elevator–acceleration vs. time for all speeds

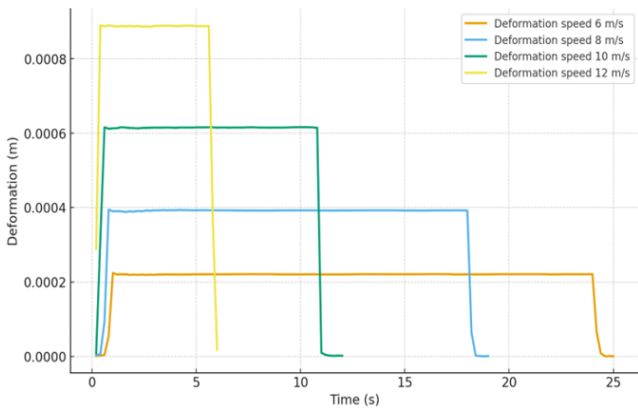


Figure 13. Dome elevator–deformation vs. time for all speeds

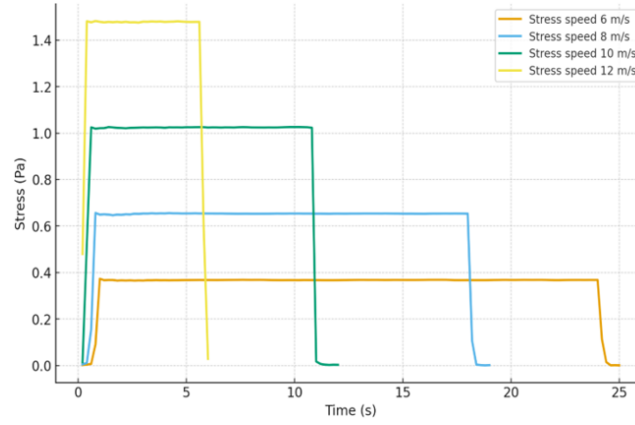


Figure 14. Dome elevator–stress vs. time for all speeds

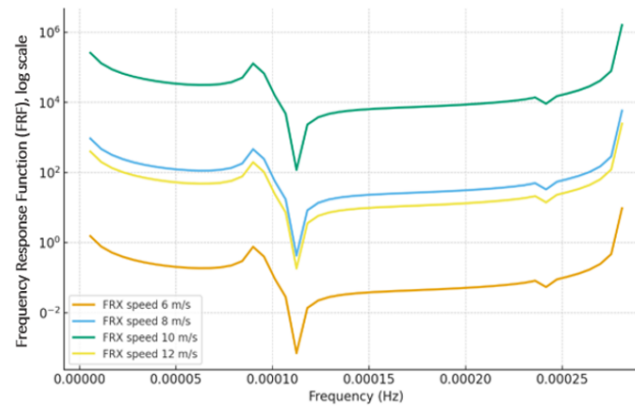


Figure 15. Dome elevator–normalized frequency response amplitude in the X-direction at different speeds

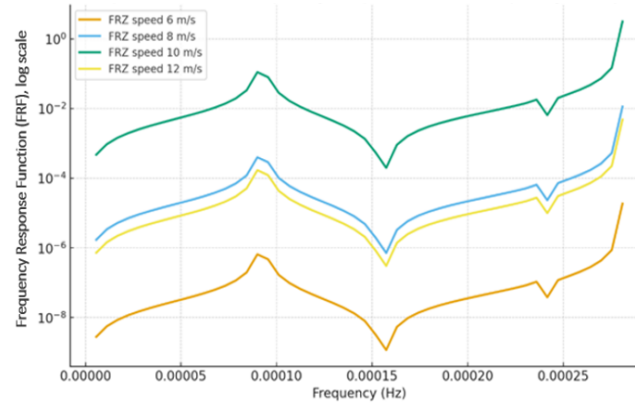


Figure 16. Dome elevator–normalized frequency response amplitude in the Z-direction at different speeds

4 Conclusion

This paper presents an integrated numerical model of air-structure interaction to capture the dynamic and aerodynamic performance of high-speed elevators accurately. This is achieved by studying changes in the shape of the elevator cabin and its speed using a dual-coupling of CFD and FEA. The following is a summary of the main findings and contributions of this paper:

- The results demonstrated that the dome shape achieves optimal aerodynamic and dynamic performance by reducing flow detachment and pressure fluctuations, thereby lowering the forces acting on the cabin.
- The results showed that the domed shape significantly reduces acceleration, deformation, and stress (approximately 41%, 35%, and 28%) while also reducing vibrations, due to improved flow efficiency.

- The results showed a critical dynamic amplification zone at an operating speed of approximately 10 m/s, where the vibrational response registers its highest values compared to other speeds. This indicates the need to consider this condition when designing damping systems and improving dynamic stability.
- The results showed, from a practical standpoint, that the domed shape of the cabin reduces air resistance and vibrations compared to other configurations. This suggests a possible improvement in system efficiency during operation.
- This study is limited to a simplified model; therefore, the results for the dome shape are a design guideline based on numerical analysis and not a definitive recommendation, as they do not include manufacturing constraints, space requirements, and design standards.

Author Contributions

Conceptualization, Z.M.A.; methodology, Z.M.A.; validation, D.A.; formal analysis, Z.M.A.; investigation, Z.M.A.; resources, M.J.A.; data curation, Z.M.A.; writing—original draft preparation, Z.M.A.; writing—review and editing, D.A. and M.J.A.; visualization, Z.M.A.; supervision, D.A.; project administration, D.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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Nomenclature

L_h	Elevator shift length (travel direction) (m)
W_h	Elevator shift width (lateral direction) (m)
H_h	Elevator shift height (m)
L	Cabin length (m)
W	Cabin width (m)
H	Cabin height (m)
t	Time variable (s)
Δt	Time step size (s)
$u(t)$	Displacement as a function of time (m)
$\dot{u}(t)$	Velocity (m/s)
$\ddot{u}(t)$	Acceleration (m/s ²)
M	Mass matrix (kg)
C	Damping matrix (N·s/m)
K	Stiffness matrix (N/m)
F_{aero}	Aerodynamic force acting on the cabin (N)
$F_{inertial}$	Inertial force (N)
E	Young’s modulus (Pa)
D	Elasticity matrix (Pa)
v	Airflow or elevator velocity (m/s)
P	Pressure (Pa)
x, y, z	Spatial coordinates (m)
u_i	Instantaneous velocity components (m/s)
k	Turbulent kinetic energy (m ² /s ²)
FRF	Frequency response function
f	Frequency (Hz)
A	Amplitude of response

Greek symbols

ρ	Density of air/material (kg/m ³)
ν	Poisson’s ratio
σ	Stress (Pa)
ϵ	Strain
ε	Turbulent dissipation rate (m ² /s ³)
μ	Dynamic viscosity (Pa·s)
α	Mass-proportional Rayleigh damping coefficient
β	Stiffness-proportional Rayleigh damping coefficient